

SUBTRACTION PUZZLES

Let's Solve the Mystery!

Grade 2 · Mathematics · Detailed Lesson Plan

Subject Mathematics	Grade 2nd Grade	Duration 60 Minutes	Theme Mystery / Puzzle Detective
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This lesson uses a fun detective-mystery theme to teach Grade 2 students the three types of subtraction puzzles — Missing Difference, Missing Subtrahend, and Missing Minuend. Students apply real-life Indian contexts (gulab jamuns, rupee coins, mangoes) to make subtraction meaningful and memorable.

1 Learning Outcomes

LO 1	Conceptual Understanding	Explain what subtraction means (taking away / finding how many are left) and identify key vocabulary.
LO 2	Vocabulary Mastery	Use the correct terms: minuend, subtrahend, difference, minus, remaining.
LO 3	Puzzle Solving — Type A	Calculate the missing difference when minuend and subtrahend are known.
LO 4	Puzzle Solving — Type B	Find the missing subtrahend using reverse thinking (subtraction or addition).
LO 5	Puzzle Solving — Type C	Determine the missing minuend by applying the addition-helps strategy.
LO 6	Strategy Application	Use number-line hops and the inverse-operation trick to verify answers.

2 Key Vocabulary

Subtraction	— The operation of taking away one number from another to find how many remain.
Minuend	— The BIG starting number in a subtraction sentence (the number we subtract FROM).
Subtrahend	— The number being taken away or subtracted.
Difference	— The result or answer of a subtraction problem — what is left.
Minus (–)	— The symbol that represents the subtraction operation.
Take Away	— A phrase meaning subtract; describes removing objects from a group.
Remaining	— What is left after subtraction has been performed.
Number Line	— A visual tool; hop BACKWARD along it to model subtraction.
Inverse Op.	— The opposite operation; addition is the inverse of subtraction.

■ Teacher Tip

Display vocabulary cards on the board throughout the lesson. Encourage students to point to the correct card whenever a term is used during discussion.

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Story Hook — The Mystery of the Missing Gulab Jamuns

Ramu made **20 delicious gulab jamuns** for his family. But when he came back, some had disappeared mysteriously! Students are invited to put on their detective hats and use subtraction to solve the mystery.

Why This Hook Works

Narrative context activates curiosity and gives subtraction a real purpose. Using a familiar Indian sweet grounds the maths in everyday life, making it more accessible for students from diverse backgrounds.

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Concept Explanation

4.1 What Is Subtraction?

Subtraction means **taking away** or finding **how many are left** after removing a quantity from a group. Every subtraction sentence has three parts.

Worked Example

$$15 - 6 = 9$$

Answer: Priya had 15 laddoos and gave 6 to Raju. 9 laddoos remain.

4.2 Parts of a Subtraction Sentence

Term	Position in Equation	Example ($18 - 5 = 13$)
Minuend	The first (large) number	18
Subtrahend	The number subtracted	5
Difference	The answer / result	13

5 The Three Puzzle Types

Type A · Missing Difference (Easiest)

Both the minuend and the subtrahend are known. Students simply perform the subtraction to find what is left.

$$18 - 5 = \underline{\quad}$$

Answer: 13

Story: Meena had 18 rotis. She served 5 to her family. How many rotis are left?

■ Teacher Tip

Have students cross out objects drawn on a whiteboard or use counters to physically model taking away. Concrete → Pictorial → Abstract progression.

Type B · Missing Subtrahend (Medium)

The minuend and the difference are known; students must figure out how much was taken away. Encourage **thinking backwards**: if $20 - ? = 13$, ask "what do I add to 13 to get 20?"

$$20 - \underline{\quad} = 13$$

Answer: 7 (because $13 + 7 = 20$)

Story: Rahul had 20 rupee coins. He spent some at the shop. Now he has 13 coins left. How many rupees did he spend?

■ **Teacher Tip**

Write the inverse equation beside the puzzle: $20 - \underline{\quad} = 13 \rightarrow 13 + \underline{\quad} = 20$. Seeing both forms helps students bridge subtraction and addition.

Type C · Missing Minuend (Most Challenging)

The subtrahend and the difference are known; students find the original starting number. The key insight is to **add the subtrahend to the difference**: $\underline{\quad} - 6 = 10$ becomes $10 + 6 = 16$.

$$\underline{\quad} - 6 = 10$$

Answer: 16 (because $10 + 6 = 16$)

Story: Sunita had some mangoes. She gave 6 away. Now she has 10. How many did she have at first?

■ **Teacher Tip**

Use the phrase "Start = What's Left + What Was Taken" as a memorable anchor. Students can say it aloud before writing the addition equation.

At-a-Glance Comparison

Puzzle Type	Equation	Strategy	Answer
Type A (Missing Difference)	$18 - 5 = \underline{\quad}$	Subtract directly; count back on number line	13
Type B (Missing Subtrahend)	$20 - \underline{\quad} = 13$	Think backwards; use addition inverse	7
Type C (Missing Minuend)	$\underline{\quad} - 6 = 10$	Add difference + subtrahend to find start	16

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Key Strategy — Reverse Thinking: Addition Helps!

When a piece is missing from a subtraction sentence, students can flip to an addition equation to find the answer. This works because addition and subtraction are **inverse operations**.

Subtraction Puzzle	→	Addition Equation	Answer
$\underline{\quad} - 6 = 10$	→	$10 + 6 = \underline{\quad}$	16
$20 - \underline{\quad} = 13$	→	$13 + \underline{\quad} = 20$	7

Real-life story: Amma bought bananas. She gave 8 to guests. 12 bananas are left. How many did she buy? → $12 + 8 = 20$ **bananas**.

7

Number Line Strategy — Hop Backward to Subtract!

The number line gives students a concrete visual model of subtraction. They start at the minuend and hop **backward** by the subtrahend, landing on the difference.

Example: Start at **15** → hop back **4** → land on **11**

$$15 - 4 = 11$$

Classroom Activity: Draw a large number line on the floor using tape (0 – 30). Students physically jump backward the correct number of spaces. Kinesthetic learners especially benefit from this embodied approach.

8 Lesson Activities (60-Minute Plan)

Activity 1 Mystery Hook & Warm-Up

■ 10 min

Read the gulab jamun story aloud with expression. Ask: "Who could have taken Ramu's gulab jamuns?" Accept creative answers to build engagement.

Quick warm-up: Count back from 20 together as a class (20, 19, 18...) clapping each number. Then pose: "If Ramu started with 20 and 7 disappeared, how many remain?" ($20 - 7 = 13$). Model on board.

Materials: Whiteboard, story prompt card

Activity 2 Vocabulary Introduction

■ 8 min

Introduce the three terms using vocabulary cards: **Minuend**, **Subtrahend**, **Difference**. Write $18 - 5 = 13$ on the board and label each part.

Check understanding: call a number, ask students to hold up the card for the correct position (e.g., "18 is the...?"). Repeat with a new equation.

Materials: Vocabulary flash cards (3 per pair)

Activity 3 Type A Puzzle Practice

■ 10 min

Model: $18 - 5 = \underline{\quad}$ (Meena's rotis story). Solve together with counters; then verify on the number line.

Independent practice: Students solve 5 Type-A problems on their worksheets using any method (counters, number line, mental maths).

Share answers; address misconceptions before moving on.

Materials: Counters, number line strip, worksheet

Activity 4 Type B & Reverse Thinking

■ 10 min

Model: $20 - \underline{\quad} = 13$ (Rahul's coins story). Demonstrate writing the inverse: $13 + \underline{\quad} = 20$. Count up from 13 to 20 on the number line.

Pair work: Partners take turns posing Type-B puzzles to each other. One partner writes the subtraction puzzle; the other flips it to addition and solves.

Materials: Mini whiteboards or worksheet, number line

Activity 5 Type C Puzzle & Addition Helps!

■ 10 min

Model: $\underline{\quad} - 6 = 10$ (Sunita's mangoes). Say aloud: "Start = Left + Taken Away". Write $10 + 6 = 16$. Check: $16 - 6 = 10$. ✓

Guided practice: Amma's banana problem together: $\underline{\quad} - 8 = 12 \rightarrow 12 + 8 = 20$. Discuss why adding gives us the start.

Challenge: Early finishers create their own Type-C story using a favourite food.

Materials: Worksheet, coloured pencils for labelling

Activity 6 Crack the Code! Puzzle Game

■ 7 min

Divide class into small groups. Each group receives a code-card with 5 subtraction puzzles (mix of Types A, B, C). Solving each puzzle reveals a letter; the letters spell a secret Indian word (e.g., MANGO or DIWALI).

First group to decode the word correctly wins a "Maths Detective" badge sticker.

Materials: Code cards, pencils, detective badge stickers

Activity 7 Quick Quiz & Wrap-Up

■ 5 min

Students solve three problems independently (one of each type):

1. $42 - 15 = \underline{\quad}$
2. $40 - \underline{\quad} = 25$
3. $\underline{\quad} - 12 = 20$

Discuss answers together. Celebrate with "You Cracked the Mystery!" — students give themselves a round of applause.

Materials: Half-sheet quiz paper

9 Guided Practice — Answer Key

Problem	Type	Working	Answer
$25 - 8 = \underline{\quad}$	Type A	Subtract: $25 - 8$	17
$35 - \underline{\quad} = 18$	Type B	$18 + \underline{\quad} = 35 \rightarrow 35 - 18$	17
$\underline{\quad} - 9 = 15$	Type C	$15 + 9 = \underline{\quad}$	24
$42 - 15 = \underline{\quad}$	Type A (Quiz)	Subtract: $42 - 15$	27
$40 - \underline{\quad} = 25$	Type B (Quiz)	$25 + \underline{\quad} = 40 \rightarrow 40 - 25$	15
$\underline{\quad} - 12 = 20$	Type C (Quiz)	$20 + 12 = \underline{\quad}$	32
$30 - \underline{\quad} = 14$	Type B (Story)	$14 + \underline{\quad} = 30 \rightarrow 30 - 14$ (Chotu's pens)	16

10 Differentiation & Inclusion

Learner Group	Strategies
Struggling Learners	Provide physical counters and a 0–30 number line strip. Pre-draw objects for story problems. Focus on Type A only; introduce B/C in a follow-up lesson.
On-Level Learners	Use all three types with story-problem context. Encourage verification by checking answers with the inverse operation.
Advanced Learners	Create their own subtraction mystery stories. Explore two-step problems (e.g., $50 - 13 - 8 = \underline{\quad}$). Attempt "Crack the Code" extension with larger numbers up to 99.
ELL / Bilingual	Use picture-based story cards. Allow mother-tongue explanations in pairs. Vocabulary cards with illustrations.

11 Assessment & Evaluation

Formative (During Lesson)

Observe student use of vocabulary cards. Check mini-whiteboard responses. Listen for correct use of terms (minuend, subtrahend, difference) during pair work. Note which students require manipulatives.

Summative (Quick Quiz)

The three-question quiz (one per type) provides a snapshot of individual mastery. Mark for: correct answer (1 pt), correct working shown (1 pt). Target: 4/6 or above = meeting expectations.

Exit Ticket

Exit Ticket Prompt

On a sticky note, complete: "Subtraction is _____. One strategy I used today was _____." Collect at the door and use to plan the next lesson.

12 Resources & Materials Checklist

Item	Quantity	Note
■ Vocabulary flash cards	1 set per pair	<i>Prepare in advance</i>
■ Counter sets (beans/coins)	1 bag per student	<i>Can use rupee coins</i>
■ Number line strips (0–30)	1 per student	<i>Print or draw on card</i>
■ Story problem worksheets	1 per student	<i>Types A, B, C mixed</i>
■ Code cards (Crack the Code)	1 per group	<i>Print & laminate</i>
■ Mini whiteboards	1 per pair	<i>Or use paper in plastic sleeve</i>
■ Maths Detective badges	1 per group	<i>Sticker or printed card</i>
■ Floor number line (tape)	1 in classroom	<i>Set up before lesson</i>

13 Extension & Homework

Homework: Ask a family member to hide some objects (e.g., spoons, chapatis). Write the subtraction mystery in your maths journal: "There were ____ objects. ____ disappeared. Now there are ____ left." Bring your mystery to class tomorrow!

Next Session Preview: More fun with subtraction puzzles and games — students will explore 2-digit subtraction with regrouping and exciting new story challenges.

■ You Cracked the Mystery! You're Maths Detectives! ■